

PsdToUIToolKit Documentation

PsdToUIToolKit is a Unity Editor tool for reading PSD/PSB layer structures and exporting them as UXML and image assets for Unity UI Toolkit.

This project currently uses Unity 2022.3.62f3. The tool runs in the Unity Editor only and does not include runtime export logic.

Features

- Open and preview PSD/PSB files.
- Inspect the PSD layer tree, layer type, layer ID, position, and size.
- Configure each layer's export state, visibility, name, and merge behavior.
- Export text layers as `ui:Label`.
- Export regular layers, group layers, or merged layers as `ui:VisualElement` elements with `background-image`.
- Replace a layer's raster export with a custom Sprite.
- Detect nine-slice images and write UI Toolkit slice styles.
- Deduplicate images within the current export and against shared image directories.
- Optionally analyze layout and export selected containers as Row, Column, or Grid layouts; absolute positioning remains the default.

Directory Structure

```
Assets/PsdToUIToolKit
├─ Editor
│   ├─ Window          # PSD UI Toolkit Editor window
│   ├─ Exporter        # Image export, UXML writing, and auto-layout analysis
│   └─ Config          # PSD settings, export path preferences, and image export
                        config
├─ PsdTools            # PSD/PSB reading and layer parsing code
├─ PSDConfig           # Export config JSON files for individual PSD files
├─ EditorConfig        # Optional global image export config JSON files
├─ Generated           # Default export directory
└─ Scenes              # Sample scenes
```

Quick Start

1. Open this project in Unity.
2. Open `Tools > PSD > UI Toolkit Editor` from the Unity menu.
3. Click `Open PSD` and select a `.psd` or `.psb` file.

4. Select a layer from the layer tree on the left, preview it in the center, and edit its export settings in the Inspector on the right.
5. Configure the export settings as needed:
 - `Image Root` : image output root directory. Default: `Assets/PsdToUIToolKit/Generated/Images` .
 - `UXML Root` : UXML output root directory. Default: `Assets/PsdToUIToolKit/Generated/Uxml` .
 - `Auto image naming` : when enabled, exported image names include the layer ID to reduce name collisions.
6. Click `Export UXML + Images` .

Default export output:

```
Assets/PsdToUIToolKit/Generated/Images/<PSD file name>/  
Assets/PsdToUIToolKit/Generated/Uxml/<PSD file name>.uxml
```

Note: each export deletes and recreates the matching image directory, such as `Generated/Images/<PSD file name>` . Do not place manually maintained assets in that directory.

Layer Settings

After selecting a layer, use the Inspector on the right to adjust its export behavior.

- `Name` : the exported element name. It also affects the image file name.
- `Export` : when disabled, the layer is excluded from export.
- `Visible` : controls whether the exported element is visible. Disabled layers are written with `display: none` in UXML.
- `Merge export` : available for group layers only. When enabled, the whole group is exported as one composited image.
- `Use custom image` : replaces the current layer's raster export with a Sprite from the project.
- `Slice / nine-slice` : attempts to detect and export the image as nine-slice.
- `Override nine-slice params` : overrides nine-slice detection parameters for the selected layer.
- `Participate local dedup` : includes the layer in duplicate detection within the current export.
- `Participate common dedup` : includes the layer in duplicate detection against shared image directories.
- `Participate in auto layout` : includes the layer in auto-layout analysis.

Auto Layout

Auto layout is an optional PSD-scoped feature and is disabled by default.

Global settings are available in the `Auto Layout (PSD)` section of the Inspector:

- `Enable auto layout` : enables layout analysis.
- `Detection mode` : analysis mode. Available values include Conservative, Balanced, and Aggressive.

- `Min confidence` : minimum confidence score. Analysis below this threshold falls back to absolute positioning.
- `Alignment tolerance` : alignment tolerance.
- `Gap tolerance` : spacing tolerance.
- `Rebuild layout tree` : rebuilds the layout tree before writing UXML. Image export behavior stays unchanged.
- `Allow virtual containers` : allows synthetic layout containers to be generated.
- `Detect background containers` : attempts to identify background containers.
- `Max nesting depth` : maximum layout nesting depth.

When auto layout is disabled, analysis fails, or confidence is too low, the exported UXML falls back to absolute positioning.

Configuration Files

PSD Export Config

Each PSD automatically gets an export config file:

```
Assets/PsdToUIToolKit/PSDConfig/<PSD file name>_uitoolkit_export_config.json
```

Layer settings are stored by layer ID. When a PSD is reopened, the tool syncs the current layer list while preserving existing layer settings.

Global Image Export Config

The following optional files live in `Assets/PsdToUIToolKit/EditorConfig`. If they do not exist, the defaults defined in code are used.

`PSD_NineSliceConfig.json` :

```
{
  "borderInset": 2,
  "pixelThreshold": 10,
  "minCenterCols": 10,
  "minCenterRows": 10,
  "minSameZone": 15
}
```

`PSD_DedupConfig.json` :

```
{
  "maeThreshold": 0.04,
  "fingerprintSize": 8
}
```

PSD_CommonDirectories.json :

```
{
  "paths": [
    "Assets/YourCommonImageFolder"
  ]
}
```

Shared image directories are scanned recursively for PNG files. Shared images referenced by UXML must ultimately be located under Unity's `Assets` directory.

Export Output

The exported UXML root node uses the PSD file name as its `name` and writes the original PSD width and height:

```
<ui:VisualElement name="psdName" style="position: relative; width: ...px; height: ...px; overflow: hidden;">
```

Regular image layers are exported as `ui:VisualElement`, and text layers are exported as `ui:Label`. By default, elements use absolute positioning. When auto layout is enabled and analysis succeeds, containers may receive UI Toolkit flex styles.

Exported images are written as PNG files and imported into Unity as Sprites. When nine-slice borders are detected, the tool writes both the Sprite border and the corresponding `-unity-slice-*` styles in UXML.

Recommendations

- Keep PSD layer names clear and stable so exported UXML element names remain readable.
- Use `Merge export` for complex groups, button backgrounds, or visual sections that should be treated as one image.
- Use `Use custom image` or shared image directory deduplication for reusable icons and common assets.
- Start with auto layout disabled, verify the absolute-positioned export, then enable auto layout gradually.
- Keep `Auto image naming` enabled when image name collisions need to be avoided.

FAQ

Why did manually added files disappear from the exported image folder?

The exporter deletes `Images/<PSD file name>` before writing the current PSD's image output. This removes stale files from previous exports. Manually maintained assets should be stored elsewhere.

Why do PSD files with the same name share config?

PSD config files are generated from the PSD file name. If two PSD files in different folders have the same name, they share the same `<PSD file name>_uitoolkit_export_config.json`. Avoid duplicate PSD names or rename the file before importing it.

Why did auto layout not take effect?

Auto layout is disabled by default and is affected by confidence, alignment tolerance, gap tolerance, nesting depth, and each layer's participation setting. When the requirements are not met, the tool falls back to absolute positioning.

Why does exported text not exactly match Photoshop?

Text layers are exported as UI Toolkit `Label` elements with basic properties such as text, font size, and color. Complex Photoshop text effects may not be fully reproduced.